[redacted]

CS1005 – Discrete Structures

Quiz 1_2

Wednesday, 4 February 2025
Time: 10 mins
Marks: 10

Note: Please write in blue or black ink. Use of pencils is not allowed. No articles (calculators etc.) can be exchanged.

1. Write the contra-positive, converse, and inverse of the following sentences.
   a. What goes around comes around. → If it goes around, then it comes around.
   b. Without petrol the car would come to a halt. → If there is no petrol, then the car halts.

(a) Contrapositive: If it doesn't come around, then it doesn't go around.
Converse: If it comes around then it goes around.
Inverse: If it doesn't go around, then it doesn't come around.

(b) Contrapos: If the car doesn't halt, then there is petrol.
Converse: If the car halts, then there is no petrol.
Inverse: ~~If the car doesn't ha~~ If there is petrol, then the car doesn't halt.

2. Using De Morgan's Law, write the negation of $7 > y \geq -4$.

$7 > y \geq -4$ can be written as $((y < 7) \land (y \geq -4))$

Its negation will be $\sim(y < 7 \land y \geq -4)$
$\equiv (y \geq 7) \lor (y < -4)$

3. Explain if the following argument is valid or not.
    a. Anyone who lives in the city Honolulu, HI also lives on the island of Oahu.
    b. Kanoe does not live in the city Honolulu, HI.
    c. Therefore, Kanoe does not live on the island of Oahu.

Let H = lives in Honolulu
Let O : lives on Oahu

(a.) $H \rightarrow O$
(b) $\sim H$
(c) $\therefore \sim O$

* Invalid argument (inverse error)